

Infania Pimentel

Human Factors Engineer · Interaction Systems · Physical UX

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Education

TUFTS UNIVERSITY, Medford, MA

May 2026

Master of Science in Human Factors Engineering

Thesis: Designed, built, and validated a wearable using glanceable visual signals to influence real-world social behavior.

UNIVERSITY OF NEW MEXICO, Albuquerque, NM

May 2021

Bachelor of Arts in Chemistry; Bachelor of Science in Psychology

Minor: Computer Science — Cum Laude

Experience

CENTER FOR ENGINEERING EDUCATION AND OUTREACH, TUFTS UNIVERSITY, Medford, MA

SociaLIGHT — Embedded Systems & Human Factors Lead (Master's Thesis)

2024–Present

- Designed a glanceable wearable system using color-based cues to support social interaction initiation among strangers.
- Defined end-to-end interaction flows across embedded hardware, transitioning from BLE/WiFi to NFC to reduce setup complexity, and used visual feedback to communicate system state at a glance.
- Conducted a two-part evaluation (usability + IRB-approved behavioral study), showing a significantly greater desire to reconnect, increased willingness to initiate interaction, and higher perceived interaction quality.
- Improved system usability across iterations, reducing setup time by 83% and achieving 100% user task completion.

Embedded Systems Engineer — Componentize Platform

2024–Present

- Developed a modular system that connects hardware and browser interfaces through reusable components, enabling live data exchange and control.
- Defined a component architecture enabling interaction across BLE, WebSocket, and serial-connected devices.
- Translated developer feedback into interaction patterns and system behavior, improving usability.
- Integrated AI-assisted tooling to support debugging and system configuration.

Embedded Systems & Human Factors Engineer — Smart Playground Initiative

2024

- Designed and implemented a proximity-based device pairing system using RSSI signal strength, replacing traditional UI selection with physical interaction.
- Developed LED-based feedback cues to communicate system state (scanning, selection, pairing) without a screen.
- Built a distributed embedded system across multiple ESP devices with defined roles and real-time communication.
- Translated user behavior and testing insights into interaction flows and system logic.

BRAY LABS MACHINE SHOP, TUFTS UNIVERSITY, Medford, MA

Machine Shop Fabrication Technician

2023–2024

- Trained students on mills, laser cutters, waterjet systems, and additive manufacturing equipment.
- Advised on manufacturability, material selection, tolerances, and machining strategies for prototype development.
- Translated CAD models into manufacturable parts, identifying design constraints early in the design process.

UNIVERSITY OF MASSACHUSETTS AMHERST, Amherst, MA

Polymer Engineering & Physics Research Assistant

Summer 2021

- Designed and fabricated experimental systems to study programmable polymer mechanics at macro scale.
- Developed Python-based 3D simulation tools to model self-assembling structures and analyze kinematics.

Skills

Embedded Systems & Hardware: ESP32 (C3/C6), Raspberry Pi, BLE, NFC, Serial Communication, Embedded Sensors, Motors, Cameras, LED Systems

Human Factors & Validation: Usability Testing, Ergonomic Analysis, IRB Design, Statistical Analysis, Risk Analysis

CAD & Fabrication: Onshape, SolidWorks, Figma, Laser Cutting, Waterjet, Additive Manufacturing, DFM, Milling

Programming: Python, MicroPython, JavaScript, HTML/CSS, WebSocket, PyScript, MATLAB, R

Systems & Interaction: System Architecture, Hardware–Software Integration, Interaction Design for Embedded Systems, Feedback System Design

Languages: English (Fluent), Spanish (Reading/Writing)